CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 8: Remainder and Factor Theorems

Topic-wise short notes with solved examples

These two theorems give the remainder of a polynomial division without dividing, and tell whether a linear expression is a factor. They are then used to factorise cubic polynomials.

1 Polynomials and division

- A **polynomial** in x is written $f(x)$. Its value at $x = a$ is $f(a)$, found by substituting a for x.
- When $f(x)$ is divided by a divisor, we get a quotient and a remainder:

$$\text{Dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$$

- When the divisor is linear (degree 1), the remainder is a constant.

2 The remainder theorem

- **Theorem**: if $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$.

Divisor	Remainder
$x - a$	$f(a)$
$x + a$	$f(-a)$
$ax - b$	$f\left(\frac{b}{a}\right)$
$ax + b$	$f\left(-\frac{b}{a}\right)$

- Rule: put the divisor equal to zero, solve for x, and substitute that value in $f(x)$.

Example 2.1 Find the remainder when $f(x) = x^3 - 3x^2 + 4x - 5$ is divided by $(x - 2)$.

SOLUTION

$x - 2 = 0$ gives $x = 2$

$f(2) = 8 - 12 + 8 - 5$

Remainder $= -1$

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Example 2.2 Find the remainder when $f(x) = 2x^3 + 3x^2 - 5x + 1$ is divided by $(2x + 1)$.

SOLUTION

$2x + 1 = 0$ gives $x = -\frac{1}{2}$

$f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{8}\right) + 3\left(\frac{1}{4}\right) + \frac{5}{2} + 1 = -\frac{1}{4} + \frac{3}{4} + \frac{5}{2} + 1$

Remainder $= 4$

3 The factor theorem

- **Theorem**: $(x - a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.
- So a divisor is a factor exactly when the remainder is zero.
- Similarly $(ax + b)$ is a factor of $f(x)$ if $f\left(-\frac{b}{a}\right) = 0$.

Example 3.1 Show that $(x - 2)$ is a factor of $x^3 - 7x + 6$.

SOLUTION

$f(2) = 8 - 14 + 6 = 0$

Since $f(2) = 0$, $(x - 2)$ is a factor.

Example 3.2 Find the value of a if $(x - 2)$ is a factor of $x^3 + ax^2 - 4x + 4$.

SOLUTION

$f(2) = 0$: $8 + 4a - 8 + 4 = 0$

$4a = -4$

$a = -1$

4 Finding unknown constants

- Each condition (a factor, or a given remainder) gives one equation. Two unknowns need two conditions.

Example 4.1 The polynomial $f(x) = x^3 + ax^2 + bx - 6$ has $(x - 1)$ as a factor and leaves remainder 12 when divided by $(x - 3)$. Find a and b.

SOLUTION

$f(1) = 0$: $1 + a + b - 6 = 0$, so $a + b = 5$

$f(3) = 12$: $27 + 9a + 3b - 6 = 12$, so $9a + 3b = -9$ and $3a + b = -3$

Subtract: $2a = -8$, so $a = -4$ and $b = 9$

$a = -4, b = 9$

5 Factorising a cubic polynomial

- **Step 1**: find one factor by trial. Try $x = 1, -1, 2, -2, 3, -3, \ldots$, which are factors of the constant term, until $f(a) = 0$. Then $(x - a)$ is a factor.
- **Step 2**: divide $f(x)$ by $(x - a)$ to get a quadratic quotient.
- **Step 3**: factorise the quadratic.

Example 5.1 Factorise $x^3 - 7x + 6$ completely.

SOLUTION

$f(1) = 1 - 7 + 6 = 0$, so $(x - 1)$ is a factor.

Dividing: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6)$

$x^2 + x - 6 = (x + 3)(x - 2)$

$x^3 - 7x + 6 = (x - 1)(x - 2)(x + 3)$

Example 5.2 Factorise $2x^3 + x^2 - 13x + 6$ completely, and hence solve $2x^3 + x^2 - 13x + 6 = 0$.

SOLUTION

$f(2) = 16 + 4 - 26 + 6 = 0$, so $(x - 2)$ is a factor.

Dividing: $2x^3 + x^2 - 13x + 6 = (x - 2)(2x^2 + 5x - 3)$

$2x^2 + 5x - 3 = (2x - 1)(x + 3)$

$(x - 2)(2x - 1)(x + 3)$; **the solutions are** $x = 2$, $x = \frac{1}{2}$ **and** $x = -3$.

REMEMBER

Quick tests: if the sum of all the coefficients is 0, then $(x - 1)$ is a factor. If the sum of the coefficients of the even powers equals the sum of the coefficients of the odd powers, then $(x + 1)$ is a factor.

6 Common mistakes

- Substituting $x = a$ for the divisor $(x + a)$. The value to use is $x = -a$.
- Sign errors with odd powers of a negative number: $(-2)^3 = -8$.
- Stopping after one factor. "Factorise completely" needs all three linear factors.
- Leaving out a missing term in long division. Write $x^3 + 0x^2 - 7x + 6$.